

Flux sectors of pyrochlore ice: log-concave counts, an octahedral winding support, and a certified failure of real-rootedness

Lyndon Drake, with Claude Code (Anthropic Claude 5 models)

2026-09-06

Abstract

The ice states of a periodic pyrochlore cell are graded by the flux vector $\mathbf{W}$ through the three coordinate cut planes, which every hexagon flip preserves, and $N(\mathbf{W})$ counts the states in one sector. Three facts are reported on six cells, from $(1, 1, 1)$ with 16 spins to $(1, 2, 3)$ with 96 spins and 3.08×10^9 ice states. The counts are log-concave with interval support along every lattice line in thirteen directions, and so are the 24 one-dimensional marginals, in exact integer arithmetic on the primitive vector of the support lattice. The support of N is exactly $2\mathbb{Z}^3$ intersected with the octahedron $a|W_x| + b|W_y| + c|W_z| \leq 4abc$, the containment proved from the ice rule and the parity of the cut-plane crossing number, the realisation of every such point checked at all six cells. And the log-concavity is not inherited from real-rootedness, since only seven of the 24 marginals have all roots real and none beyond degree eight. The counts come from two independent exact enumerators whose whole histograms agree entry by entry wherever both run, and the counts of distinct real roots of all 24 marginals are certified in Isabelle/HOL by the Archive of Formal Proofs implementation of Sturm's theorem, independently of the Python computation they agree with. For thirteen of the marginals, the seven that are real-rooted and six of the failures, the verdict itself is certified; for the other eleven the step from the count to the verdict rests on a squarefreeness check that Isabelle did not finish and Python supplies.

1 The model and the flux grading

Pyrochlore spin ice puts a spin on each bond of the diamond lattice, whose sites are the centres of the corner-sharing tetrahedra, and admits the configurations in which exactly two of the four arrows at every site point inwards (Bramwell and Gingras 2001; Castelnovo, Moessner, and Sondhi 2008). An (a, b, c) block of conventional cubic cells with periodic boundaries holds $8abc$ sites and $16abc$ spins. Positions are integers in units of $a/4$, so the

box is $(4a, 4b, 4c)$ and each of the four A-to-B nearest-neighbour displacements has all components ± 1 .

The smallest ice-preserving move reverses the six arrows around a hexagonal plaquette, and is available only when they already run head to tail, which makes the resulting chain the Rokhsar–Kivelson Hamiltonian of quantum spin ice at its RK point (Rokhsar and Kivelson 1988; Henley 2004; Hermele, Fisher, and Balents 2004). What matters here is its conservation law. A hexagon bounding a face of the infinite lattice is contractible, so reversing it moves no net flux across any cut plane, and the flux vector

$$\mathbf{W} = (W_x, W_y, W_z), \quad W_j = \frac{1}{4c_j} \sum_b \sigma_b \operatorname{sgn}(d_{bj}),$$

is constant along every trajectory. Here $\sigma_b = \pm 1$ is the arrow variable of bond b , d_{bj} the j th component of its A-to-B displacement, and $(c_x, c_y, c_z) = (a, b, c)$; the divisor turns a polarisation sum into a flux, since a bond spans a quarter of a cubic edge along each axis. A second routine, counting the signed arrows across the single plane $x = 1/2$, agrees with that formula on sampled states at $(1, 1, 1)$ and $(1, 1, 2)$ (`tests/test_enumerate.py`, `test_winding_by_polarisation_matches_cut_plane`). The ice manifold is therefore graded, and the object of this note is the integer histogram $N(\mathbf{W})$, which in the Coulomb phase is the distribution of the emergent gauge field's flux (Henley 2010).

2 Counting a sector exactly

Two enumerators produce N , sharing no code path. The first is a frontier dynamic program over a bond schedule, whose key records the ice-rule occupancy of the sites still open at the current cut together with the running polarisation sums, with Python integers as values. Nothing is materialised, so cost follows the peak number of live keys rather than the number of states. The second packs a state into one 64-bit word, walks the search tree depth-first, and reads the flux off by two population counts per axis against precomputed masks. Where both run, the whole histogram is compared entry by entry rather than just the totals, and they agree everywhere; the exception is $(1, 2, 3)$, where 96 spins do not fit the packed word and the enumerator refuses rather than truncating.

Table 1: Cells, exact counts, and the two-method gate; all figures from experiments/04_lorentzian_giant/results/results.json.

cell	spins	ice states	$N(\mathbf{o})$	sectors	DP peak	
					keys	agree
(1, 1, 1)	16	90	12	25	72	yes
(1, 1, 2)	32	3,618	388	69	368	yes
(1, 1, 3)	48	181,122	15,024	137	904	yes
(1, 1, 4)	64	9,765,378	637,988	229	1,680	yes
(1, 2, 2)	64	2,891,562	221,628	217	42,160	yes
(1, 2, 3)	96	3,076,306,362	196,465,480	453	123,432	n/a

The cubic (2, 2, 2) cell, the smallest with genuinely cubic geometry, is out of reach, since a bounded probe reached 8,661,950 live keys at 4.216 GB by step 41 of 128, projecting a peak of 7.1×10^8 keys and 347 GB.

3 The winding support

Statement. For the (a, b, c) cell,

$$\text{supp } N = 2\mathbb{Z}^3 \cap \{ \mathbf{w} : a|W_x| + b|W_y| + c|W_z| \leq 4abc \}.$$

The containment $\subseteq$ is proved. For parity, the bonds crossing the plane $x = 1/2$ number $4bc$, since $16abc$ bonds are shared equally among the $4a$ half-integer planes perpendicular to x , the count recorded in `icemix.frozen.winding_support`. If P of them carry their arrow in the positive sense and Q in the negative then $W_x = P - Q = 4bc - 2Q$, which is even, and likewise on the other axes.

The octahedron comes from the ice rule one tetrahedron at a time. The four A-to-B displacements $(1, 1, 1), (1, -1, -1), (-1, 1, -1), (-1, -1, 1)$ sum to zero, so at an A site with two arrows out and two in the polarisation moment $\sum_b \sigma_b \mathbf{d}_b$ is twice the sum of the two outward displacements. Each of the six such pairs sums to ± 2 along one coordinate axis, so every A site contributes exactly $\pm 4\mathbf{e}_k$ for a single k . Every bond has exactly one A end, so summing over the $4abc$ A sites writes the polarisation vector $(4aW_x, 4bW_y, 4cW_z)$

as a sum of $4abc$ vectors of ℓ^1 norm 4, whence $4a|W_x| + 4b|W_y| + 4c|W_z| \leq 16abc$. Setting two components to zero gives $|W_x| \leq 4bc$, $|W_y| \leq 4ac$ and $|W_z| \leq 4ab$; each of those is attained at all six cells, and the bound itself is pinned by `tests/test_enumerate.py`, `test_winding_range_is_bounded_by_the_cut_size`.

The reverse containment is a check rather than a proof. At each of the six cells the support of the computed histogram equals the set above, which is what fixes the sector counts of Table 1 at 25, 69, 137, 229, 217 and 453 (`tests/test_frozen.py`, `test_winding_support_is_2z3_in_the_octahedron`). That every admissible lattice point is realised on every cell is a conjecture with six supporting cases. One consequence is needed below: the lattice of differences of the support is exactly $2\mathbb{Z}^3$, of index 8 in $\mathbb{Z}^3$, since parity puts it inside and each $\pm 2\mathbf{e}_j$ lies in the octahedron.

4 Log-concavity

A finite integer sequence $n_0, \dots, n_m$ read along consecutive points of a line is *interval-supported* when its nonzero entries are contiguous and *log-concave* when $n_k^2 \geq n_{k-1}n_{k+1}$ at every interior k ; the *discriminant* of the line is the exact integer $\min_k(n_k^2 - n_{k-1}n_{k+1})$ over interior k .

Because the support lies in $2\mathbb{Z}^3$, the point $\mathbf{W} \pm \mathbf{e}_j$ is not a lattice point, so the literal inequality holds as $0 \geq 0$ everywhere and every axis line appears riddled with holes, which makes a pass and a fail read off it equally worthless. The test therefore has to be stated on the lattice itself, which is computed first by integer row reduction, each direction $\mathbf{v}$ going in on its primitive lattice vector, the least positive multiple $t\mathbf{v}$ that the lattice contains, and that is $2\mathbf{v}$ throughout. The thirteen directions are the three axes, the six face diagonals $\mathbf{e}_j \pm \mathbf{e}_k$ and the four body diagonals $\mathbf{e}_x \pm \mathbf{e}_y \pm \mathbf{e}_z$, and every maximal line of the support is tested, not only those through the origin.

Result. At all six cells, in all thirteen directions, on every line, the sequence is interval-supported and log-concave, with the minimum discriminants of Table 2. The single qualification is that at $(1, 1, 1)$ every body-diagonal line meets the support in at most two points, leaving no interior term, which is recorded as *vacuous* rather than as a pass.

Table 2: Minimum discriminants of the joint histogram over all lines in each class, and the symmetry reading. ‘Symmetry’ counts the signed axis permutations preserving the cell extents that N obeys, over those the shape admits.

cell	symmetry	$\mathbf{W} \rightarrow -\mathbf{W}$	axis	face diagonal	body diagonal
(1, 1, 1)	48/48	yes	52	3	no interior
(1, 1, 2)	16/16	yes	160	10	138
(1, 1, 3)	16/16	yes	324	21	56,184
(1, 1, 4)	16/16	yes	544	36	15,251,160
(1, 2, 2)	16/16	yes	592	10	523,840
(1, 2, 3)	8/8	yes	1,272	5,112	2,790

The symmetry column is measured rather than assumed and comes out complete; inversion holds because complementing every arrow maps the ice manifold to itself and negates the flux. Four marginals are taken at each cell, the three W_j and the diagonal flux $s = W_x + W_y + W_z$, and all 24 are log-concave with interval support on a support of step 2. The smallest cell shows how tight that can be: each W_j marginal at (1, 1, 1) is 1, 16, 56, 16, 1 with discriminant 200, while the s marginal is 9, 24, 24, 24, 9, flat across its interior and log-concave with equality. Two negative controls guard the reading, a planted diagonal violation and a planted internal zero, each shown to be caught. The scope is finite: six cells, up to 96 spins, and no theorem for general (a, b, c) . What it rules out is an artefact of the smallest cell, the property surviving a cell with over three billion states and 453 sectors.

5 Real-rootedness, and why it would have mattered

The oldest reason for a log-concave combinatorial sequence is a real-rooted generating polynomial. If $p(u) = \sum_k a_k u^k$ has non-negative coefficients and only real roots, Newton’s inequalities give $a_k^2 / \binom{n}{k}^2 \geq (a_{k-1} / \binom{n}{k-1})(a_{k+1} / \binom{n}{k+1})$, which is strictly stronger than log-concavity and also forbids internal zeros (Brändén 2014). Had the marginals been real-rooted, the previous section would have been a corollary.

Each step-2 marginal is read as the coefficients of a polynomial in $u = t^2$, a change of

variable that leaves log-concavity alone. The number of distinct real roots follows from Sturm's theorem over the rationals, using the canonical sequence $p, p', -\text{rem}(p, p')$ and so on evaluated at $\pm\infty$; the number of distinct complex roots is $\deg p - \deg \gcd(p, p')$; the polynomial is real-rooted exactly when the two agree. All arithmetic is in exact fractions.

Result. Of the 24 marginals, 7 are real-rooted and 17 are not. By degree, 6 of the 7 marginals of degree 4, 1 of the 6 of degree 8, and none of the 4 at degree 12, the 5 at degree 16 or the 2 at degree 24. The real-rooted seven are W_x, W_y, W_z at $(1, 1, 1)$, and W_z at $(1, 1, 2), (1, 1, 3), (1, 1, 4)$ and $(1, 2, 3)$.

The convenient summary 'real-rooted only up to degree four' has an exception in each direction, since the s marginal at $(1, 1, 1)$ has degree 4 and only 2 real roots while W_z at $(1, 2, 3)$ has degree 8 and 8 real roots. What holds without qualification is that no marginal of degree 12 or more is real-rooted. Newton's inequalities are therefore not the route to Table 2, and whatever accounts for the log-concavity of these counts has to survive genuinely complex roots.

6 The Isabelle/HOL certification

That verdict rests on a few dozen lines of hand-written exact arithmetic in the `real_root_counts` routine of `icemix/logconcave.py`, so all 24 cases were recomputed in Isabelle/HOL 2025-2 against the Archive of Formal Proofs entry `Sturm_Sequences` (Eberl 2014, 2015), an independent implementation of the same classical theorem. Sturm's theorem gives the number of *distinct* real roots in an interval as the drop in sign changes of the canonical chain between the endpoints, and the AFP entry formalises it and exposes the proof method `sturm`. Each polynomial receives `degree_*` ($\deg p = n$, by `eval`, all 24), `sturm_*` ($\text{card}\{x \in \mathbb{R} : p(x) = 0\} = k$, by `sturm`, all 24 including the two of degree 24), `sqfree_*` ($\gcd(p, p') = 1$ over $\mathbb{C}[x]$, by `eval`, 13 of 24) and a `verdict_*`.

The verdicts take three forms, and the distinction is the substance of the formalisation. The 7 real-rooted cases assert $\text{card}\{x : p(x) = 0\} = \deg p$. Six of the 17 failures get something stronger than a count, namely some $z \in \mathbb{C}$ with $p(z) = 0$ and $\text{Im } z \neq 0$, from a general lemma proved once (`Squarefree_Nonreal.nonreal_root_exists`): a square-free complex polynomial has exactly $\deg p$ distinct roots by the fundamental theorem of algebra, and the real roots inject into those with $\text{Im } z = 0$, so a deficit forces a root off

the real axis. Without squarefreeness the deficit could be a repeated real root instead. The remaining 11, all of degree 12 or more, carry only $\text{card}\{x : p(x) = 0\} < \deg p$, since deciding $\gcd(p, p') = 1$ by `eval` runs an executable subresultant-style gcd that proved squarefreeness for every degree-8 case in 5 to 11 seconds, including one with a coefficient of 2,218,493,976, and timed out beyond 240 seconds on every case of degree 12 or more, including one whose largest coefficient is 56,222. That wall is degree rather than coefficient size, and `sturm` itself met no such wall at any degree tested.

All 24 Isabelle root counts agree with the Python ones, and the agreement means something because Isabelle proves each count itself while the Python value only states the goal, so a mismatch would appear as a failed build. Three things remain trusted: the coefficients, checked only by the two enumerators against each other; the `eval` steps, which reflect into the code generator and so rest on a wider base than a kernel proof; and the generator that transcribes coefficient lists into the theory text, a step visible in the theories and diffable against the results file. Rebuilding needs the AFP session heap once, then the session:

```
isabelle build -b -o threads=6 Sturm_Sequences
isabelle build -d formal/isabelle -o threads=6 IceMixingSturm
```

The theories themselves are never edited by hand; `python3 formal/isabelle/gen_theory.py` regenerates them from the results file.

The recorded run of the second command took 10 minutes 8 seconds elapsed and 21 minutes 3 seconds of CPU on 6 threads, exiting 0 with no error lines.

7 What stands

Three things are not established here: any theorem for general (a, b, c) , or for a cubic cell beyond $(1, 1, 1)$; any proof that every admissible lattice point carries a state; and any explanation of the log-concavity beyond the elimination of its obvious candidate. A related question is settled elsewhere, in the negative. Log-concavity along lines is the sort of shadow a Lorentzian or negatively dependent structure would cast (Brändén and Huh 2020; Anari et al. 2024), which makes it natural to ask whether the bond indicators of the uniform ice measure are negatively correlated in some diagonal sign gauge. They are

not, and the obstruction is an unbalanced triangle on three bonds of the 16-spin cell, so no larger cell can repair it (docs/reports/2026-09-04-stage4abc.md, §3).

8 Related work

Log-concavity and unimodality of combinatorial sequences have a long catalogue, surveyed by Stanley (1989), updated by Brenti (1994), and treated from unimodality through log-concavity to real-rootedness by Brändén (2014), where Newton’s inequalities are the classical bridge from real roots to ultra-log-concavity. The strongest recent framework is that of Lorentzian polynomials (Brändén and Huh 2020), whose coefficient arrays are log-concave along lines in a normalised sense and whose sampling consequences drive the matroid results of Anari et al. (2024). The present histogram is a candidate object for that theory, and the facts here bear on it in opposite directions, the coefficients behaving as predicted while the pairwise negative dependence that usually accompanies such behaviour fails.

Flux sectors are standard equipment in the statistical mechanics of constrained models. In the Coulomb phase the flux is the coarse-grained emergent field with a Gaussian distribution at large size (Henley 2010), a picture made quantitative for spin ice by Jaubert et al. (2013). The dimer analogue is the height function and its tilt, where counting becomes the surface tension of the variational principle of Cohn, Kenyon, and Propp (2001), strictly convex on the interior of the Newton polygon by Kenyon, Okounkov, and Sheffield (2006), with Kenyon (2009) the accessible account. What is reported here is the finite-cell, exact-integer shadow of the concavity in tilt those results establish asymptotically in two dimensions, with the octahedron playing the part of the Newton polygon.

9 Appendix: Artefacts

Paths are relative to the repository root.

path	contents
icemix/lattice.py, icemix/enumerate.py	the cell and the winding signs; frontier_count, enumerate_ice_states, winding, winding_by_cut_plane
icemix/logconcave.py	support lattice, primitive vector, line and marginal reports, real_root_counts
icemix/frozen.py	winding_support and the octahedron argument
experiments/04_lorentzian_giant/results/summary.json	every result in Tables 1 and 2, under items.4a
formal/isabelle/	ROOT, Squarefree_Nonreal.thy, six generated Marginals_*.thy, gen_theory.py, summary.json, final-build.log
tests/test_frozen.py, tests/test_stage4abc.py	the winding-support test, the real-root pins, the log-concavity tests and their negative controls
docs/reports/2026-09-04- stage4abc.md,	the stage-4a report with negative dependence in §3; the 24-row certification
docs/reports/2026-09-04-isabelle- sturm.md	table and the squarefreeness scope cut

References

- Anari, Nima, et al. (2024), ‘Log-Concave Polynomials II: High-Dimensional Walks and an FPRAS for Counting Bases of a Matroid’, *Annals of Mathematics*, 199/1: 259–99, <https://doi.org/10.4007/annals.2024.199.1.4>.
- Bramwell, Steven T., and Michel J. P. Gingras (2001), ‘Spin Ice State in Frustrated Magnetic Pyrochlore Materials’, *Science*, 294: 1495–501.

- Brändén, Petter (2014), *Unimodality, Log-Concavity, Real-Rootedness and Beyond*.
- Brändén, Petter, and June Huh (2020), ‘Lorentzian Polynomials’, *Annals of Mathematics*, 192/3: 821–91, <https://doi.org/10.4007/annals.2020.192.3.4>.
- Brenti, Francesco (1994), ‘Log-Concave and Unimodal Sequences in Algebra, Combinatorics, and Geometry: An Update’, in *Jerusalem Combinatorics ’93*, Contemporary Mathematics (Providence, RI), 178:71–89, <https://doi.org/10.1090/conm/178/01893>.
- Castelnovo, Claudio, Roderich Moessner, and Shivaji L. Sondhi (2008), ‘Magnetic Monopoles in Spin Ice’, *Nature*, 451: 42–5.
- Cohn, Henry, Richard Kenyon, and James Propp (2001), ‘A Variational Principle for Domino Tilings’, *Journal of the American Mathematical Society*, 14/2: 297–346, <https://doi.org/10.1090/S0894-0347-00-00355-6>.
- Eberl, Manuel (2014), ‘Sturm’s Theorem’, *Archive of Formal Proofs* (January).
- Eberl, Manuel (2015), ‘A Decision Procedure for Univariate Real Polynomials in Isabelle/HOL’, *Proceedings of the 2015 Conference on Certified Programs and Proofs* 75–83, <https://doi.org/10.1145/2676724.2693166>.
- Henley, Christopher L. (2004), ‘From Classical to Quantum Dynamics at Rokhsar-Kivelson Points’, *Journal of Physics: Condensed Matter*, 16: S891–8.
- Henley, Christopher L. (2010), ‘The “Coulomb Phase” in Frustrated Systems’, *Annual Review of Condensed Matter Physics*, 1: 179–210, <https://doi.org/10.1146/annurev-conmatphys-070909-104138>.
- Hermele, Michael, Matthew P. A. Fisher, and Leon Balents (2004), ‘Pyrochlore Photons: The $U(1)$ Spin Liquid in a $S=1/2$ Three-Dimensional Frustrated Magnet’, *Physical Review B*,

69: 064404.

Jaubert, L. D. C., et al. (2013), ‘Topological-Sector Fluctuations and Curie-Law Crossover in Spin Ice’, *Physical Review X*, 3/1: 011014, <https://doi.org/10.1103/PhysRevX.3.011014>.

Kenyon, Richard (2009), *Lectures on Dimers*.

Kenyon, Richard, Andrei Okounkov, and Scott Sheffield (2006), ‘Dimers and Amoebae’, *Annals of Mathematics*, 163/3: 1019–56, <https://doi.org/10.4007/annals.2006.163.1019>.

Rokhsar, Daniel S., and Steven A. Kivelson (1988), ‘Superconductivity and the Quantum Hard-Core Dimer Gas’, *Physical Review Letters*, 61: 2376–9.

Stanley, Richard P. (1989), ‘Log-Concave and Unimodal Sequences in Algebra, Combinatorics, and Geometry’, *Annals of the New York Academy of Sciences*, 576/1: 500–35, <https://doi.org/10.1111/j.1749-6632.1989.tb16434.x>.